1. Name of the Student

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2. Title

Timer-Based LED Flasher PCB (Assignment Question 6)

Design overview

A signage company needs a simple circuit that periodically flashes an LED to attract attention on information displays and warning signs. The design uses an NE555 timer in astable mode driving an LED through a current-limiting resistor, powered from a 9 V supply via a 2-pin connector. The board was designed in KiCad (schematic, footprint assignment, layout and routing).

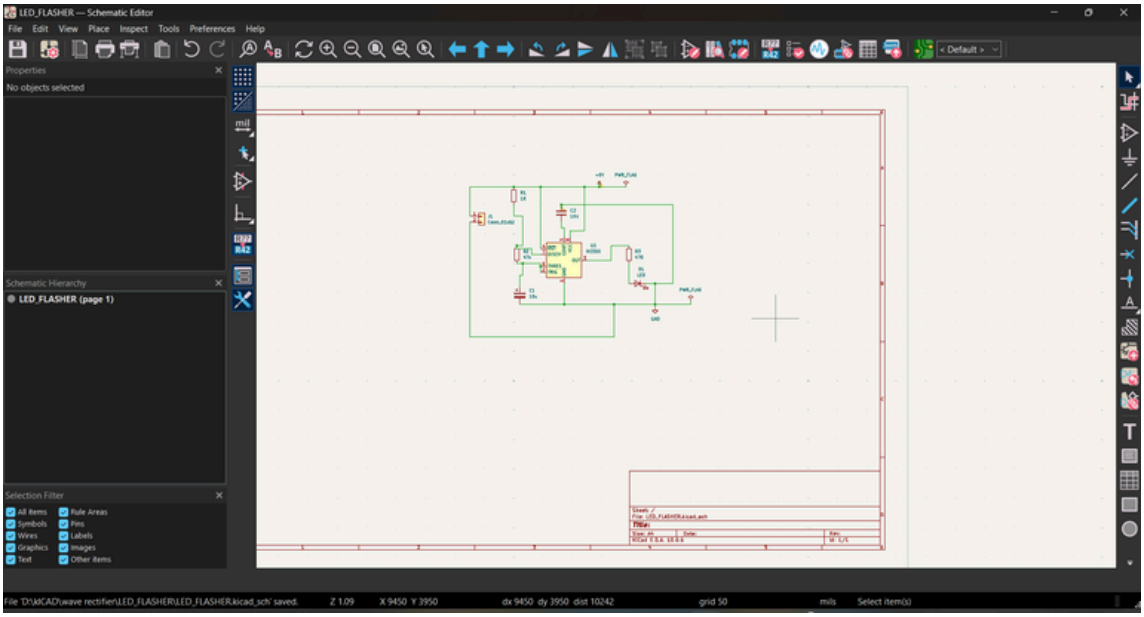
Frequency calculation

$f = 1.44 / ((R1 + 2 \cdot R2) \cdot C1) = 1.44 / ((1\ k\Omega + 2 \times 47\ k\Omega) \times 10\ \mu F) = 1.44 / 0.95 \approx$ **1.5 Hz**

High time $t_H = 0.693\ (R1 + R2)\ C1 \approx 0.33$ s; low time $t_L = 0.693\ R2\ C1 \approx 0.33$ s; duty cycle ≈ 51%.

LED current: $I = (V_{OUT} - V_F) / R3\ (9 - 1.5 - 2.0) / 470 \approx 12$ mA (safe for a standard 5 mm LED).

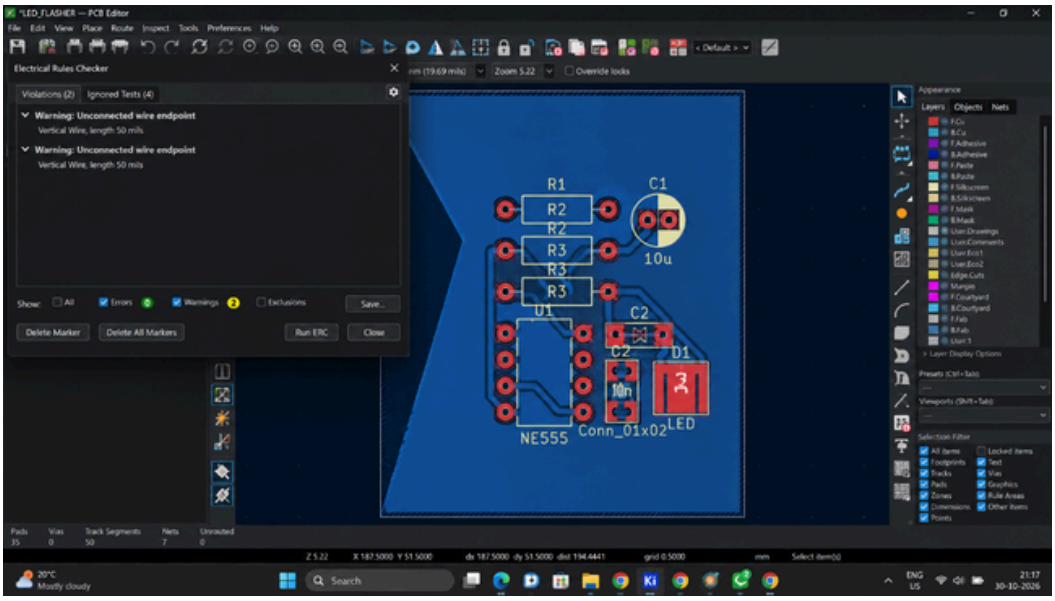
3. Schematic



Netlist / connections

Net	Connections
+9V	J1-1, U1 pin 8 (VCC), U1 pin 4 (RESET), R1
GND	J1-2, U1 pin 1, C1(-), C2, D1 cathode
N_DISCH	R1, R2, U1 pin 7 (DISCH)
N_TIMING	R2, U1 pin 6 (THRES), U1 pin 2 (TRIG), C1(+)
N_CTRL	U1 pin 5 (CTRL), C2
N_OUT	U1 pin 3 (OUT), R3
N_LED	R3, D1 anode

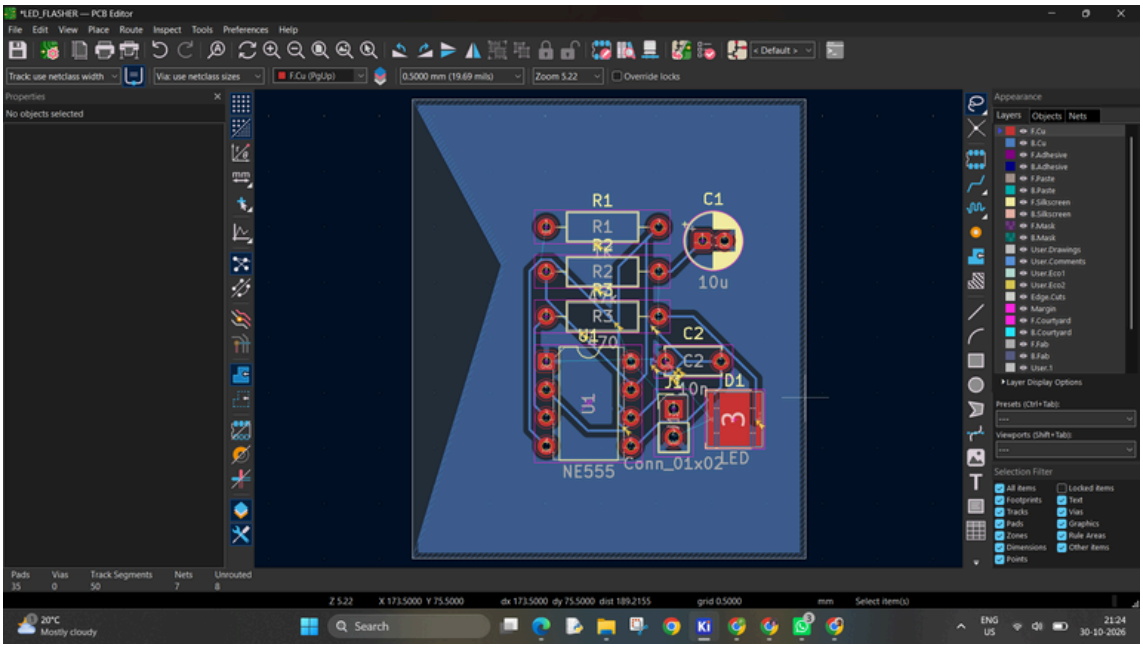
4. Screenshot of Footprint After Routing Is Completed



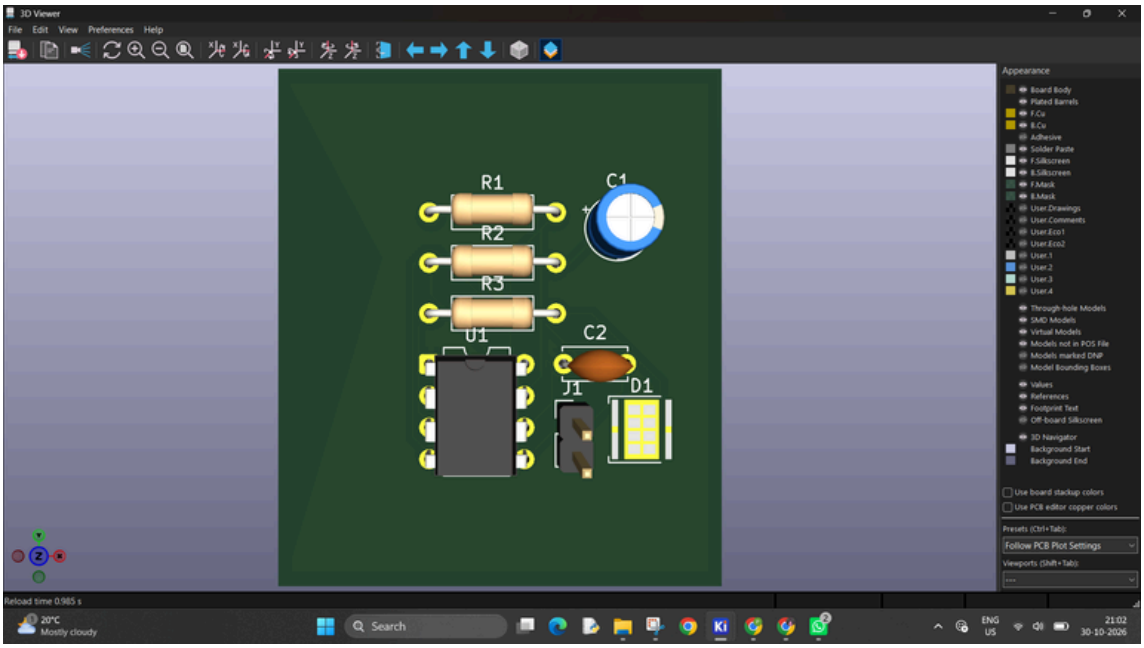
Footprint assignment

Ref	Footprint
U1	Package_DIP:DIP-8_W7.62mm
R1, R2, R3	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
C1	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
C2	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
D1	LED_THT:LED_D5.0mm
J1	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical

5. 2D PCB View


*LED_FLASHER — PCB Editor
File Edit View Place Route Inspect Tools Preferences Help
R1
C1
10u
R2
R3
C2
D1
NE555
Conn_01x02
LED

6. 3D PCB View


3D Viewer
File Edit View Preferences Help
R1
R2
R3
C1
C2
U1
J1
D1
Appearance
Search

7. Bill of Materials (BOM)

Ref	Qty	Value	Footprint	Description
U1	1	NE555	DIP-8_W7.62mm	555 timer IC
R1	1	1 k	R_Axial_DIN0207 P10.16	Timing resistor
R2	1	47 k	R_Axial_DIN0207 P10.16	Timing resistor
R3	1	470	R_Axial_DIN0207 P10.16	LED current limiting
C1	1	10 uF	CP_Radial_D5.0mm_P2.00mm	Timing capacitor (electrolytic)
C2	1	10 nF	C_Disc_D5.0mm_P5.00mm	Control-pin bypass
D1	1	LED red	LED_D5.0mm	Flasher LED
J1	1	Conn_01x02	PinHeader_1x02_P2.54mm	9 V power input

Total components: 9 (8 line items).